INTE 22283 – Mobile Application Development

Homework Activity

General Instructions

1. Develop all applications using **Android Studio**.
2. Design attractive and user-friendly interfaces using appropriate Android layouts.
3. Validate all user inputs before processing.
4. Use meaningful variable names and follow proper coding standards.
5. Add comments where necessary to improve code readability.
6. Each application must be developed as a separate Android Studio project.
7. Submit:
    - A PDF report (3–5 pages) containing:
        - Introduction
        - Application screenshots
        - Implementation details
        - Challenges faced
        - Conclusion

Question 1 – Movie Ticket Booking App (10 Marks)

Scenario

A cinema plans to introduce a simple mobile application that allows customers to preview their movie ticket booking details before confirming the reservation.

Develop an Android application that satisfies the following requirements.

Functional Requirements

1. Design the **Movie Ticket Booking** screen using appropriate Android layouts.
2. The application should contain the following input fields:
    - Customer Name
    - Movie Name (Spinner)
    - Show Time (Spinner)
    - Number of Tickets (EditText)

3. Include the following buttons:
    - **Calculate Total**
    - **View Booking Summary**
4. When the **Calculate Total** button is clicked:
    - Validate all user inputs.
    - Calculate the total ticket price using the following rate:
        - **Ticket Price = LKR 1,200 per ticket**
    - Display the calculated total on the same screen.
5. When the **View Booking Summary** button is clicked:
    - Pass the following information to a second activity using an **Explicit Intent**:
        - Customer Name
        - Movie Name
        - Show Time
        - Number of Tickets
        - Total Amount
6. The second activity should display a booking confirmation page showing all received information in a well-formatted layout.
7. Include a **Back** button to return to the booking screen.

Android Concepts Covered

- Android Layouts
- Views (TextView, EditText, Spinner, Button)
- Event Handling
- Input Validation
- Explicit Intent
- Passing Data Between Activities
- Receiving Data from Intent

Question 2 – Restaurant Menu App (10 Marks)

Scenario

A local restaurant requires a mobile application to present its menu and basic business information using Google's Material Design guidelines.

Develop an Android application that satisfies the following requirements.

Functional Requirements

1. Design the application using **Material Design** principles.
2. The Home screen should include:
    - Material Toolbar
    - Restaurant Logo
    - Welcome Message
3. Display at least **three food categories** using Material Cards:
    - Rice
    - Burgers
    - Beverages
4. Implement **Bottom Navigation** (or Navigation Drawer) with the following pages:
    - Home
    - Menu
    - Contact
5. Each page should display appropriate sample content.
6. Apply a consistent Material Design theme, including suitable colors, typography, icons, and spacing.
7. Include at least one Material Button on the application.

Android Concepts Covered

- Material Design
- Material Components
- Toolbar
- Material Cards
- Material Buttons
- Bottom Navigation or Navigation Drawer
- Navigation Between Screens

Question 3 – Student Directory App (10 Marks)

Scenario

A university requires a simple mobile application to display a directory of students using RecyclerView.

Develop an Android application that satisfies the following requirements.

Functional Requirements

1. Display a list of at least **10 students** using **RecyclerView**.
2. Each student item should display:
    - Student Name
    - Registration Number
    - Department
3. Design a custom item layout using **CardView**.
4. Implement:
    - Student Model Class
    - RecyclerView Adapter
    - ViewHolder
5. Store sample student information using an **ArrayList**.
6. When a student item is clicked, display the student's name using a **Toast** message.

Android Concepts Covered

- RecyclerView
- Adapter
- ViewHolder
- CardView
- Model Class
- ArrayList
- Click Events

Submission Requirements

Each submission must include:

- PDF Report

- Screenshots of the application's user interfaces
- Source code with proper comments